

Generation Interconnection System Impact Study Report

For Queue Project Q21-1698

Lynn Valley 100 kV

August 16, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1698. The Interconnection Customer proposes a 49.0 MW battery energy storage facility interconnecting at the Lynn Valley 100 kV substation in the Illinois area. The request was received on March 23, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 503 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 14 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$7,283,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Pueblo Xtra 230/100 kV Transformer Bank	Flambeau Hydroelectric Station 100/1kV Auto Bank 1	145.2	156	5.3
Price Hill to Pueblo Xtra 100 KV	Flambeau Hydroelectric Station 100 kV/1 kV Autotransformer T1	142.5	155	5.3
Flambeau Hydroelectric Station to Pueblo Xtra 230 KV L	Oxbow to North Bonneville 100 KV L	125.2	512	7.6
Fushing to Lynn Vly 100 KV	Talentino Sw Sta to Indian Springs 100 KV	123.5	124	66.5
Lnn Vly to Talentino Sw Sta 100 KV	Camel Tacks to Flushing 100 KV	121.1	122	60.0
Talentino Sw Sta to Indian Springs 100 KV	Front Range to Shoreline 100 KV	119.4	118	14.5
Anza to Ashlan Ave 230 KV	Shoreline to Front Range 100 KV	114.8	457	31.6
Dededo to Dairymans 100 KV line	Front Range to Shoreline 100 KV	110.6	179	28.6
Ashlan Ave to Carlyss 230 KV Line	Shorline to Front Range 100 KV	109.9	464	31.6
Moore Rd to Dededo Substation 100 KV L	Front Range to Shoreline 100 KV Line	109.3	190	28.6
Camel Tracks to Flushing 100 KV Line	Front Range to Shoreline Substation 100 KV L	106.2	131	33.6
Guardian Substation to Welpport 100 KV	Front Range to Shoreline 100 KV	105.8	132	7.8
Sewall to Flushing 100 KV line	Cajalco Tap to Rio Linda 100 KV L	104.4	106	20.6
Abilene Mulberry Crk to Coram 100 KV	Anza to Ashlan Ave 230 KV L	104.1	136	24.3

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Pueblo Xtra 230/100 kV Transformer Bank	Replace transformer with higher-rated unit	\$10,442,000	1% (\$82,000)
Price Hill to Pueblo Xtra 100 KV	Rebuild line with higher-capacity conductor	\$11,612,000	1% (\$91,000)
Flambeau Hydroelectric Station to Pueblo Xtra 230 KV	Rebuild line with higher-capacity conductor	\$19,511,000	0% (\$83,000)
Fushing to Lynn Vly 100 KV	Rebuild line with higher-capacity conductor	\$4,775,000	37% (\$1,772,000)
Talentino Sw Sta Substation to Lynn Valley 100 KV	Rebuild line with higher-capacity conductor	\$6,403,000	22% (\$1,396,000)
Talentino Sw Sta to Indian Springs 100 KV	Rebuild line with higher-capacity conductor	\$10,086,000	5% (\$463,000)
Anza to Ashlan Ave 230 KV	Rebuild line with higher-capacity conductor	\$24,495,000	3% (\$669,000)
Dededo to Dairymans 100 KV line	Rebuild line with higher-capacity conductor	\$19,052,000	8% (\$1,458,000)
Ashlan Ave to Carlyss 230 KV Line	Reconductor line	\$4,092,000	3% (\$112,000)
Moore Rd to Dededo 100 KV	Reconductor line	\$1,088,000	8% (\$83,000)
Camel Tracks Substation to Flushing 100 KV	Reconductor line	\$3,607,000	14% (\$485,000)
Guardian Substation to Welpport 100 KV	Reconductor line	\$2,812,000	2% (\$65,000)
Flushing to Sewall 100 KV	Reconductor line	\$2,715,000	12% (\$324,000)
Abilene Mulberry Crk to Coram 100 KV Line	Reconductor line	\$2,285,000	9% (\$200,000)
	Total allocated to project		\$7,283,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.